

Wildfire Evacuation Equity

AUDITING WILDFIRE EVACUATION IN CALIFORNIA

TRACK 1: ACCELERATING EQUITABLE EVACUATION

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The gap between a fire being seen and the order to leave

The Question

“How long does it take from a fire being seen to an evacuation order, and does that gap increase for vulnerable communities?”

~ 50 Mins

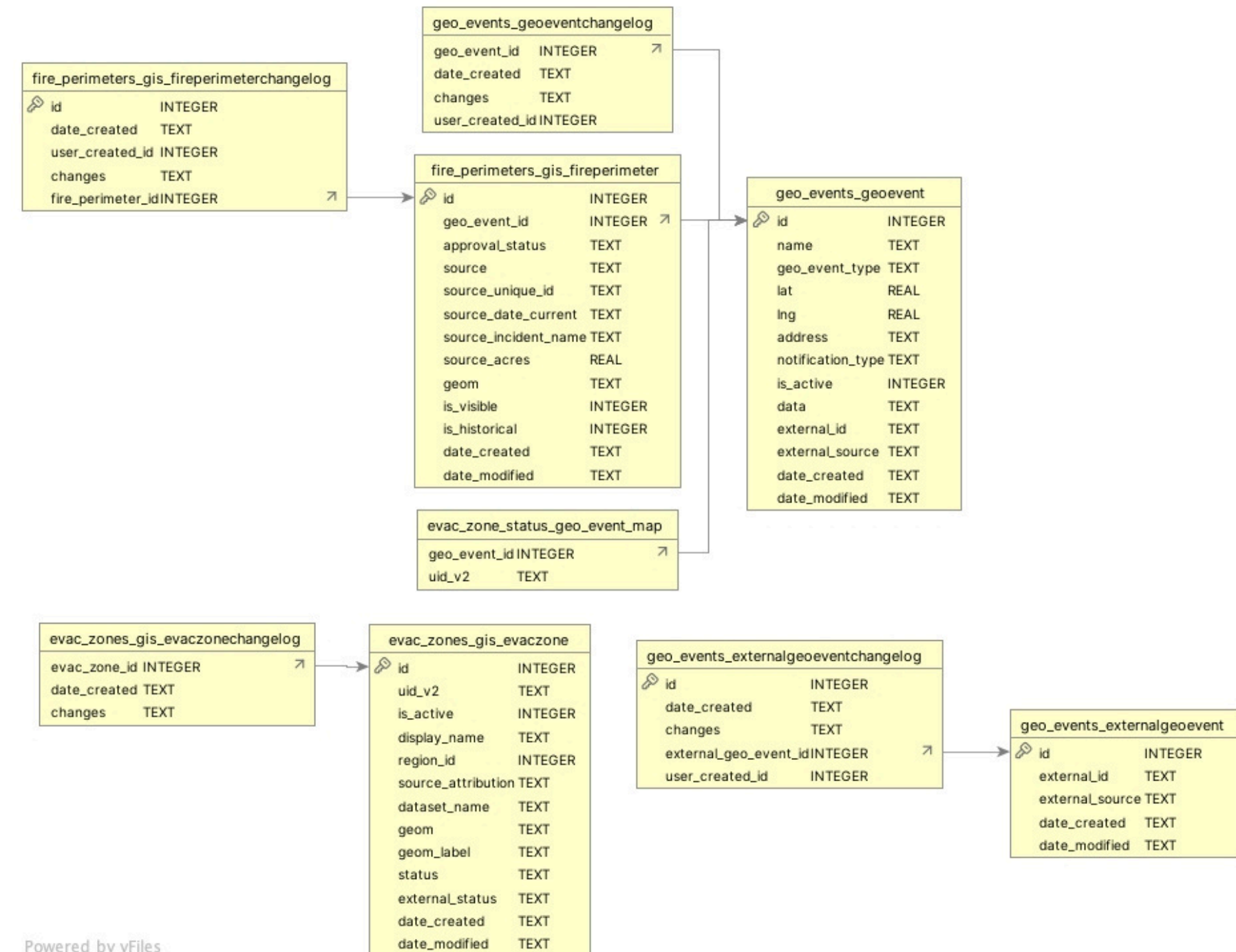
- median time from first sighting to evacuation order

Datasets & Schema

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WatchDuty data - 9 datasets

- geo_events: fire records (name, location, time, etc.)
- geo_events_changelog: reporter updates used to calculate alert lag
- evac_zones + changelog: evacuation zone status over time
- fire_perimeters + changelog: fire boundary changes over time
- external_geo_event_changelog: external source updates
- evac_zone_status_geo_event_map: links zones to fire events

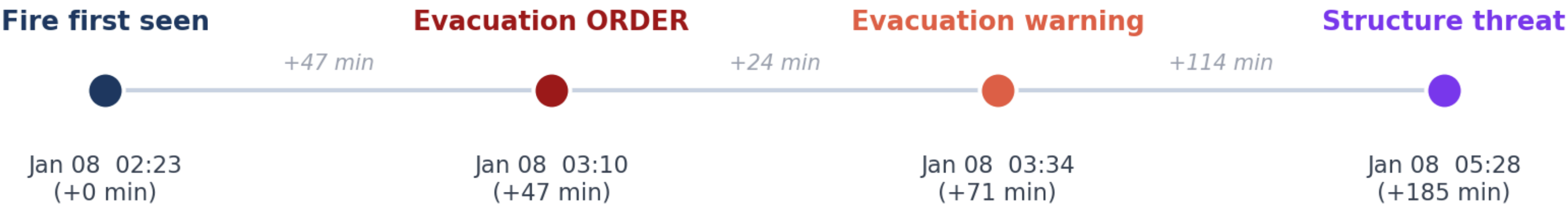


Census data - CDC SVI 2022

- 9,109 California census tracts
- 6 indicators used to calculate the vulnerability scores of each county: no internet, no vehicle, aged 65+, limited English, disability, below poverty (150%)

2025 Los Angeles Fire: Eaton Fire Case

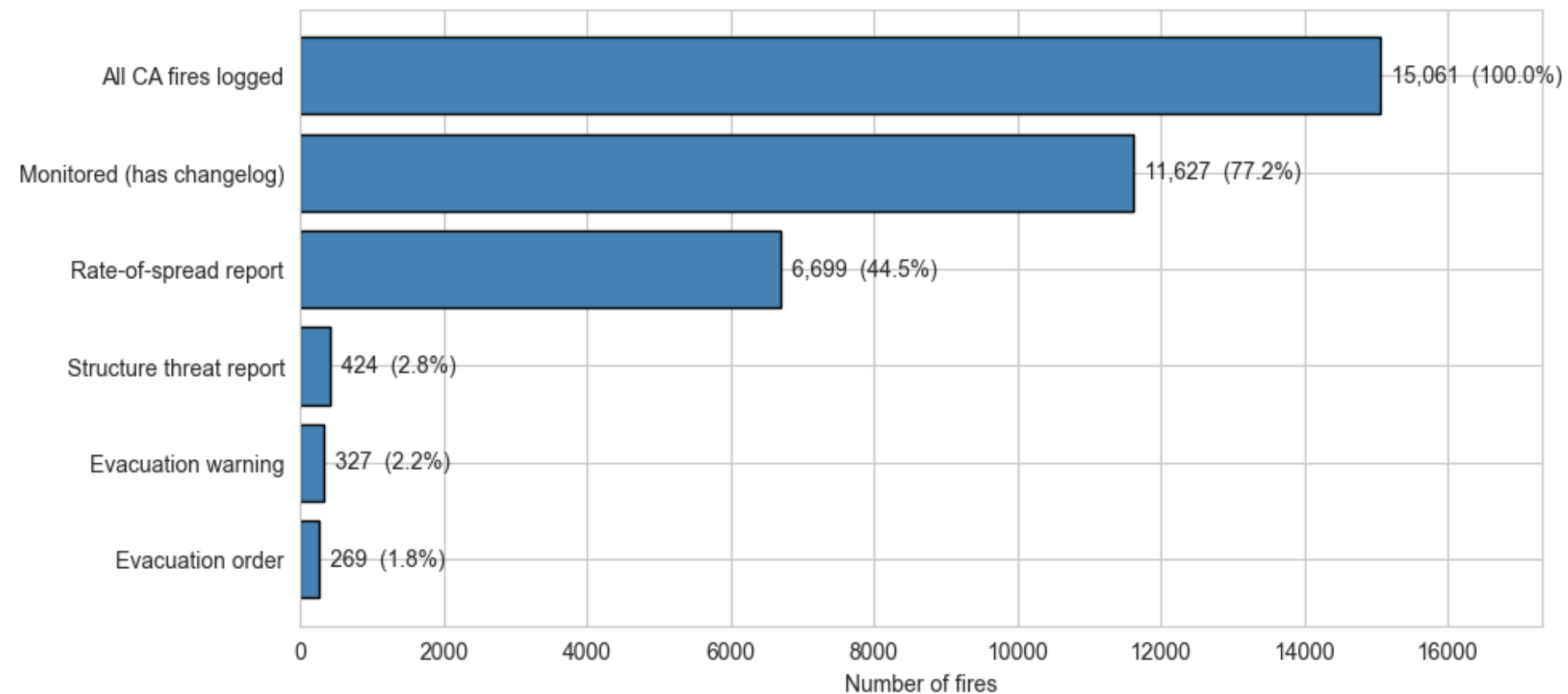
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Evacuation orders went out 47 minutes after the fire was first recorded.

Most fires are never escalated to evacuation order

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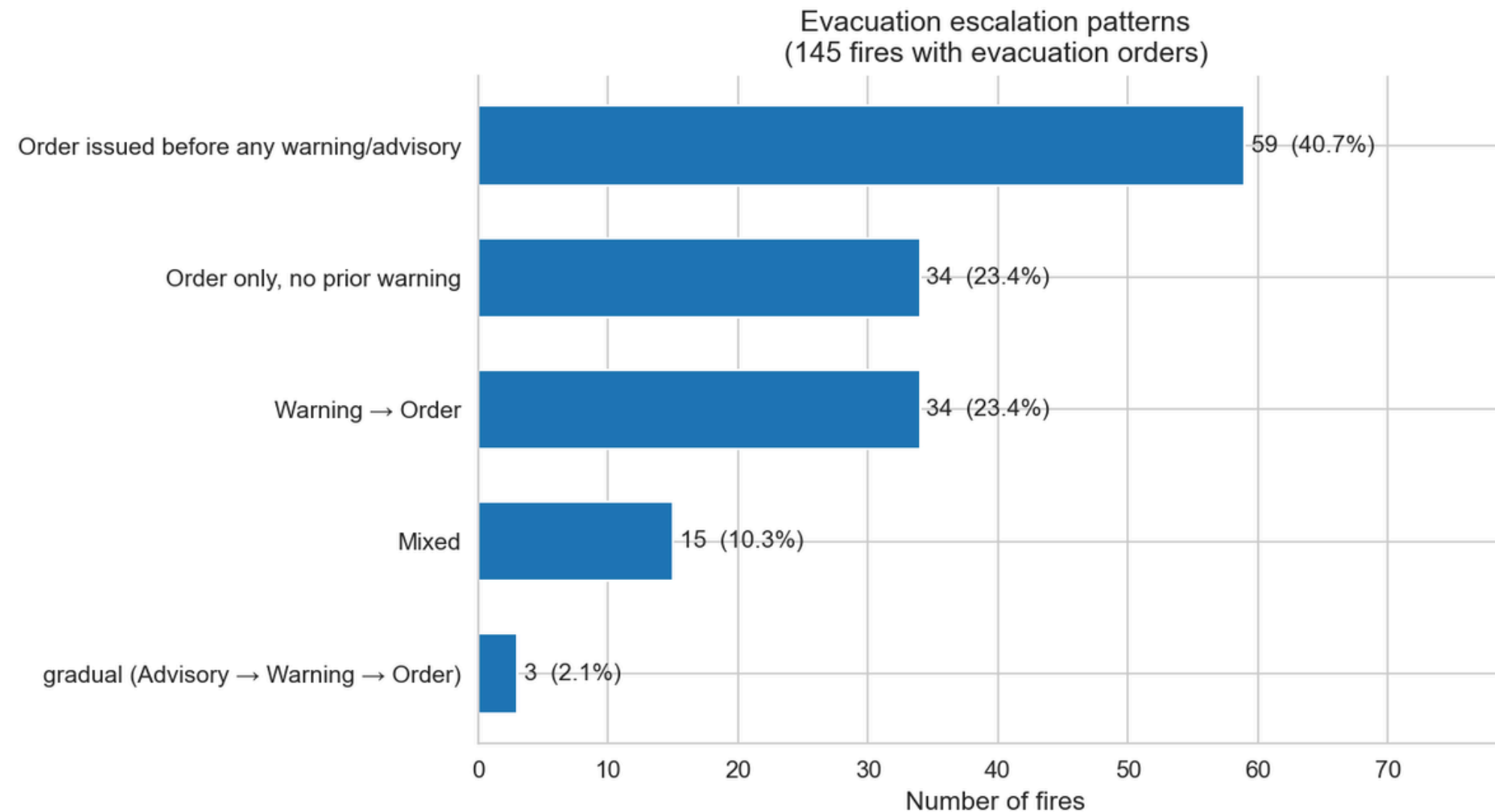


- ~15,000 CA wildfires
- ~11,600 monitored
- **ONLY ~250 with evacuation orders**

Only 1.8% of fires ever reach evacuation order,
which makes every minute of lag in those cases count

Evacuation Order Pattern

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~ **2.1 %**

Advisory → Warning → Evacuation

~ **64%**

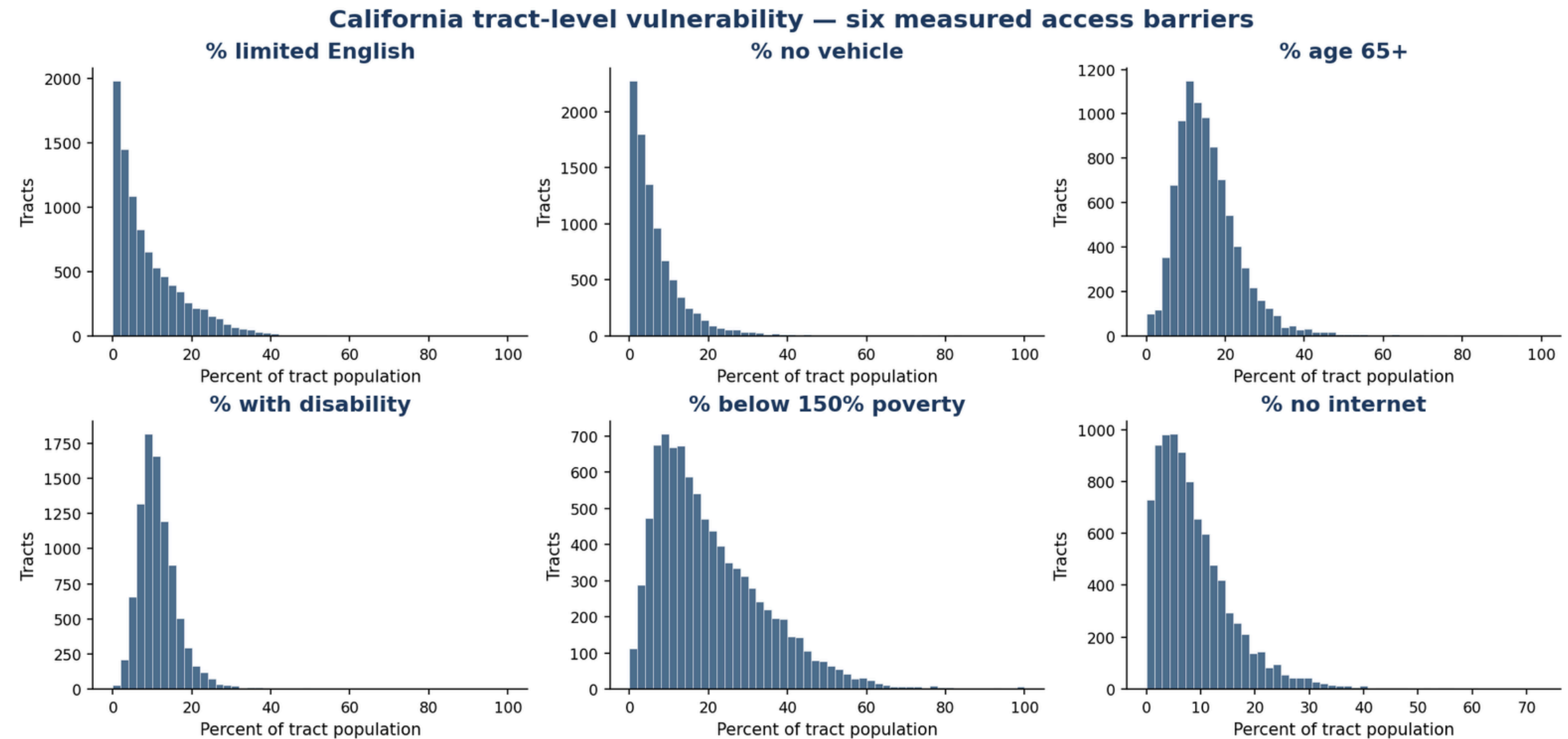
Straight to Evacuation without prior notice

Vulnerability Distribution Across Tracts

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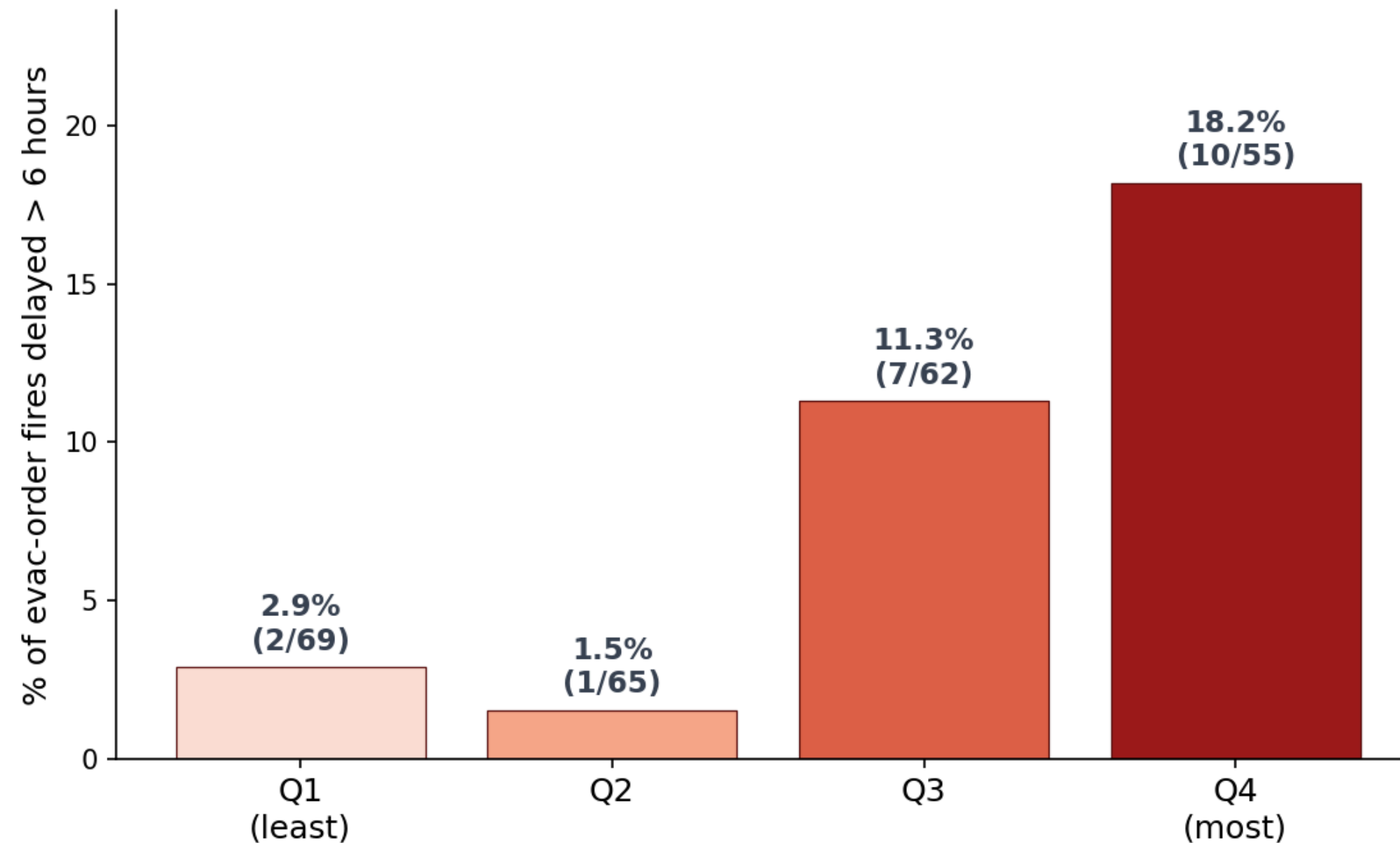
Measured barriers

- Limited English
- No vehicle in the household
- Age 65+
- Disability
- Below 150% poverty
- No internet at home



Delay in Different Vulnerability

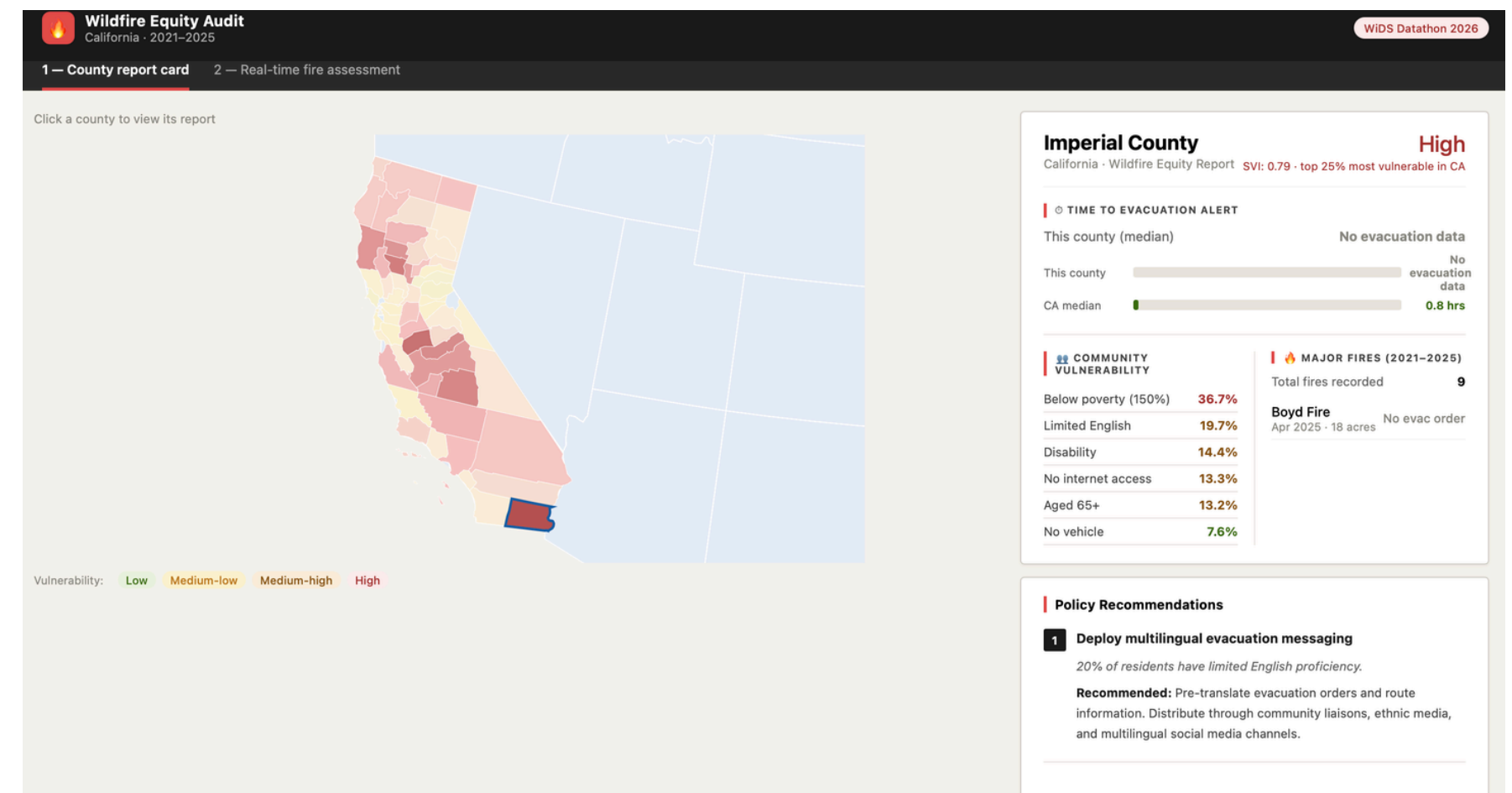
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In the most vulnerable counties,
fires are 6× more likely to face delays over 6 hours.

Our Solution: Wildfire Equity Dashboard for county emergency managers

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Tab 1: County Report Card

- Select any California county
- See alert lag, community vulnerability breakdown, and major fires history
- Get data-driven policy recommendations

Our Solution: Wildfire Equity Dashboard for county emergency managers

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Tab 2: Real-time Fire Assessment

- Input county, visual severity, and time
- Get a recommended alert level based on historical CA fire data
- Adjusts for community vulnerability and night fires

 **Wildfire Equity Audit**
California · 2021–2025

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1 — County report card2 — Real-time fire assessment

Real-time fire assessment
Enter fire details to get an alert recommendation based on historical data.

County *

Glenn County

Visual severity *

☐ Minor — small, appears contained

☒ Moderate — growing, smoke visible

☐ Major — large flames, rapid spread

☐ Extreme — out of control

Time of report *

09/08/2025

21:00 PM

Assess this fire →

Glenn County
Vulnerability: **High**
Based on CA historical fire data (2021–2025)

RECOMMENDED ALERT LEVEL

Immediate evac order
Base level: Evacuation advisory → escalated

Extreme fire behavior. Historical data: 44.7% of extreme fires require evacuation. Issue order immediately.

↑ Escalated +1: SVI 0.67 ≥ 0.60 — high-vulnerability county historically shows longer alert lags (Section 10d).

↑ Escalated +1: Night fire in high-vulnerability county — compounded access barriers (no internet, no vehicle, limited English).

ESTIMATED FIRE SIZE

10–100 acres
Based on moderate severity fires in CA

PREDICTED AFFECTED GROUPS

Aged 65+	16.9%
No internet	13.7%
Disability	12.7%
Limited English	8.0%
No vehicle	3.6%

 **Night fire detected**

Fire reported at hour 21:00. Night fires in high-vulnerability counties are automatically escalated one alert level.

Thank You



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